

SOLAR RADIATION ANALYSIS

STACWARS_AKS-3

__PS-2

1.Executive Summary

This report analyses the supplied solar-observation dataset, creates time-series and environmental visualizations, summarizes daily solar activity, and trains a Random Forest regression model to estimate solar radiation.
Numerical findings below are calculated by running the included Python script against the supplied CSV.

PS2 SOLAR RADIATION ANALYSIS — SUMMARY
Valid observations: 32,686
Date range: 2016-09-01 00:00:08 to 2016-12-31 23:55:01
Mean radiation: 207.1247
Maximum radiation: 1601.2600
Mean-radiation peak hour: 12:00
Strongest absolute linear correlation: Temperature (0.7350)
Temperature correlation: 0.7350
Humidity correlation: -0.2262
Global peak date/time: 2016-09-04 / 2016-09-04 12:15:04
Peak radiation: 1601.2600
Temperature at peak: 61
Humidity at peak: 93
Wind speed at peak: 9.0Sunrise: 2016-09-04 06:08:00; sunset: 2016-09-04 18:35:00; peak after sunrise: 0 days 06:07:04

Top 5 days by average radiation:

Date	DailyMaxRadiation	DailyAverageRadiation	ObservationCount
2016-09-19	1085.75	335.504400	275
2016-09-02	1062.01	333.471219	279
2016-09-22	1053.17	325.515880	284
2016-09-25	1075.29	319.921514	284
2016-09-28	1029.30	312.400390	282

2.Required Dependencies:

Package	*Purpose*
i: pandas	Read CSV, parse timestamps , group daily observations.
ii: numpy	Numerical calculations and cylindrical time features.
iii: matplotlib	Create and save plots.

3.How To Run The Code:

1.Download or copy the code file from the repository shared by AKS-3 .

2.Paste the code to the ***google colab*** coding section.(Google colab is highly recommended for this code testing as the coding part is done exclusively in the google colab section by team AKS-3.)

3.Before the final run of the code for checking ,ensure that the dataset provide in the PS-2 is uploaded in the google colab notebook where the code was pasted.
(NOTE: The dataset must have the location "/SolarPrediction.csv" in the google colab coding section as the code has used this location for analysis of the dataset.)

4.Once the above steps are successfully completed ,the code is ready for the final run.

5.At last, press the run button to check the working of the code.

4.Analysis Method And Visualisations:

Plot 1 — Radiation over time:

The script resamples radiation to hourly means before plotting. This makes the overall daily/seasonal pattern easier to read than plotting every raw observation.

Plot 2 and Plot 3— Radiation vs temperature and humidity:

Scatter plots show whether radiation tends to rise or fall as temperature and humidity change. Scatter does not prove causation.

Correlations among Radiation , Temperature and Humidity:

	Radiation	Temperature	Humidity
Radiation	1.000	0.735	-0.226
Temperature	0.735	1.000	-0.285
Humidity	-0.226	-0.285	1.000

Correlations:

A. near 1:strong positive linear relationship.

B. near 0: weak linear relationship.

C. near -1: strong negative linear relationship.

ANSWERS TO ASKED QUESTIONS IN THE PS-2 (i.e. Drawing interferences) :

Ans1:It can be seen from the plot and dataset that generally the Radiation is highest around the time 12:00:00 noon.

Ans2: From the correlation matrix , it is clear that the temperature has the strongest positive relationship with radiation.

Ans3: It can be observed that the temperature and radiation are directly proportional to each other whereas the humidity is inversely proportional to both of the above parameters.

Ans4: The plots and correlation matrix suggest that the solar radiation dataset may contain some unusual observations and outliers. Solar radiation can show sharp fluctuations over time , which may be caused by changing weather conditions, such as cloud cover, or sudden changes in sunlight intensity.
(i.e. the radiation is very high during mid days and almost zero during the night time as per the dataset given.)